Chapter route

How can multiple guests remain isolated while preserving instruction semantics, device interfaces, and resource bounds?

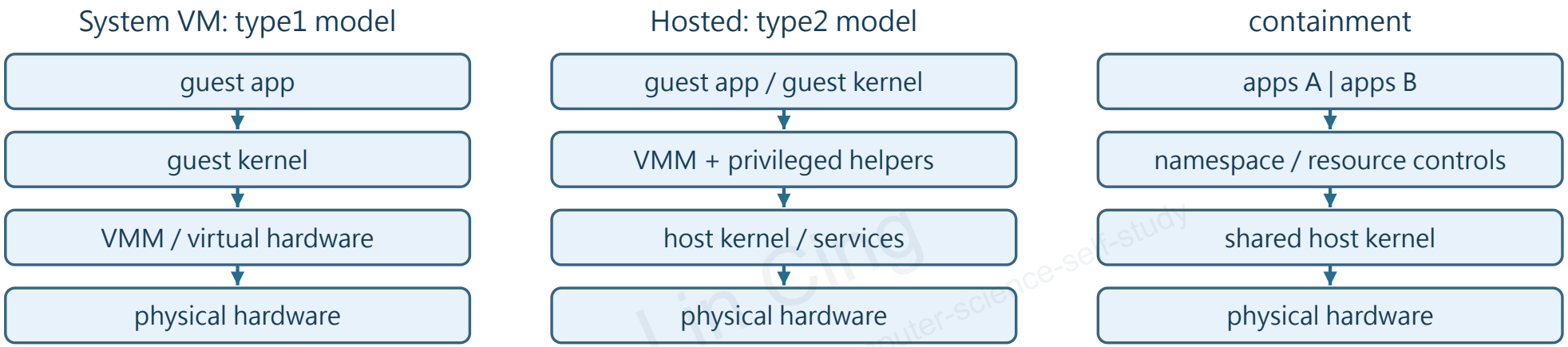
Prerequisites are OS structure, traps, address translation, and I/O.

- Distinguish type 1 and type 2 hypervisors.
- Analyze trap-and-emulate and binary translation.
- Explain memory/I/O virtualization safety and costs.

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Host, guest, and VMM boundaries

Resource-control layers differ: VMs and shared-kernel containers



The VMM/hypervisor manages physical resources and implements virtual interfaces. Guest applications call their guest OS, which controls virtual CPUs/devices. A guest syscall need not immediately enter the host kernel.

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Host, guest, and VMM boundaries (continued)

A vCPU retains guest registers, PC, state, and virtual interrupts until scheduled on a pCPU. Two VMs can both name physical page5 while second-stage mappings select different host frames. The VMM isolates shared CPUs, memory, devices, and management interfaces.

System VMs, process VMs such as JVM, cross-ISA emulators, and OS containment expose different abstractions. Identify the provided interface, kernel ownership, and physical-resource controller rather than relying on a type number.

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Motivation and design goals

IBM VM/370 in the 1970s presented separate physical-disk regions as guest minidisks for multiple OSs on one mainframe. The abstraction preserves ranges/isolation rather than exposing unrestricted disk registers.

Fidelity preserves specified guest-visible behavior except explicit differences. Resource control prevents guests seizing VMM-controlled resources. Efficiency permits most safe work to execute directly. Neither goal promises a fixed slowdown or identical timing for every workload.

Reliable interception of sensitive instructions needing mediation is a common sufficient condition for traditional trap-and-emulate. Otherwise use translation, paravirtualization, or hardware assistance. Full software emulation can interpret an ISA without dedicated VM instructions; hardware changes technique/efficiency, not the logical possibility of emulation.

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Isolation and shared failure

VMs support OS development, multiversion testing, and low-utilization server consolidation. Hypervisor/device-emulator vulnerabilities or a shared failed host can still affect several VMs. CPU, memory, I/O quotas are needed against interference; separate addresses alone do not prevent slow neighbors.

An eight-core host reserves one for VMM. A/B/C peak at 3/3/2 cores, totaling8>available7. Average demand of one each cannot guarantee no queueing at simultaneous peaks. Overcommit depends on workloads/service goals and creates no compute capacity.

Shared networks/disks introduce trust boundaries. Management can clone, snapshot, read disk images, making its authority part of the TCB even when guests are isolated.

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Suspend, snapshots, clones, and backups

Suspend retains resumable CPU/RAM/device state; a disk snapshot retains block versions at a point; a clone creates an independently runnable copy. Copying active disk files need not capture one consistent instant or matching RAM/disk state. Use shutdown, supported consistent snapshots, or guest application quiescence.

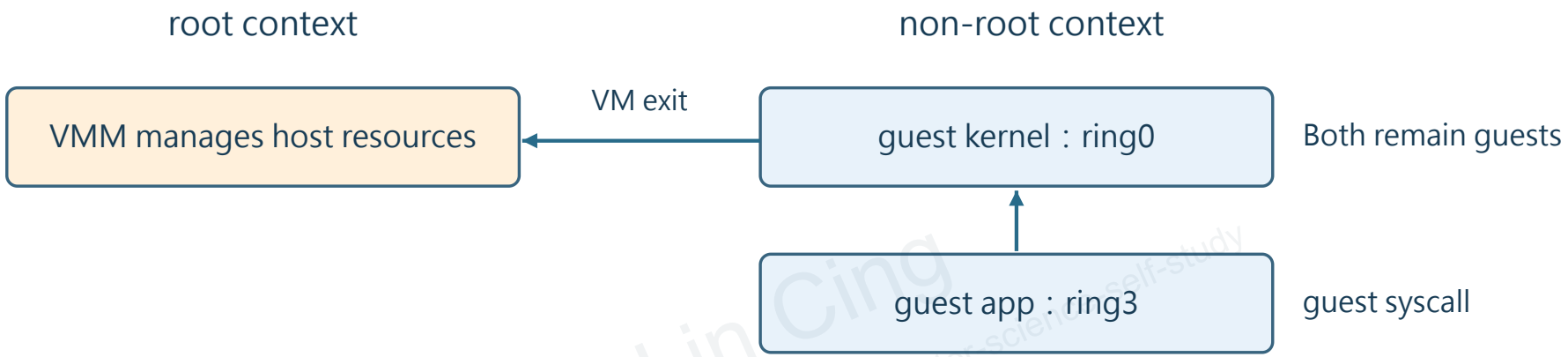
Crash consistency resembles recoverable power-loss disk state; application consistency additionally coordinates flush/transactions. External mail, payments, databases outside the snapshot are not undone by rollback. Clones must address duplicate IP/MAC, machine identities, and credentials.

Snapshots sharing the source's failure domain are not independent backups. Templates replicate both configurations and embedded vulnerabilities/credentials. Cloud/VDI APIs retain these isolation/external-state constraints.

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Virtual modes versus physical modes

User/kernel and root/non-root are separate privilege axes



Hardware-assisted model: non-root ring0 ≠ host root. Selected events exit; not every guest syscall.

In a traditional model without a separate VM mode, the VMM holds physical privilege while guest kernel/user execute with reduced physical rights. VMM retains virtual guest mode. A guest-kernel privileged operation traps, is checked against virtual authority, and is emulated. An illegal guest-user operation must produce a guest exception rather than gain authority by entering VMM.

Virtual modes versus physical modes (continued)

When a guest kernel disables interrupts, VMM clears vCPU.virtualIF and delays that guest's maskable interrupt injection. It must not permanently disable host interrupts serving other guests/devices. Re-enabling permits pending events according to the virtual architecture while retaining host control.

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Sensitive instructions that do not trap

An instruction may silently ignore some changes at reduced physical privilege, leaving a guest kernel's expected flags wrong without notifying VMM. Historical x86 POPF behavior under particular flags/privileges illustrates the gap. Lack of trap does not imply harmlessness.

Binary translation reads guest blocks, replaces mediated operations with safe equivalent VMM behavior, and caches executable translations. It must preserve indirect branches, self-modifying code, page-permission changes, and exception-visible state. Changed source code invalidates stale translations.

Paravirtualization instead changes a cooperative guest to use hypercalls. It can address similar problems with different guest ABI/modification requirements. Hardware assistance does not remove scheduling, device-emulation, and I/O costs.

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Hardware-assisted execution

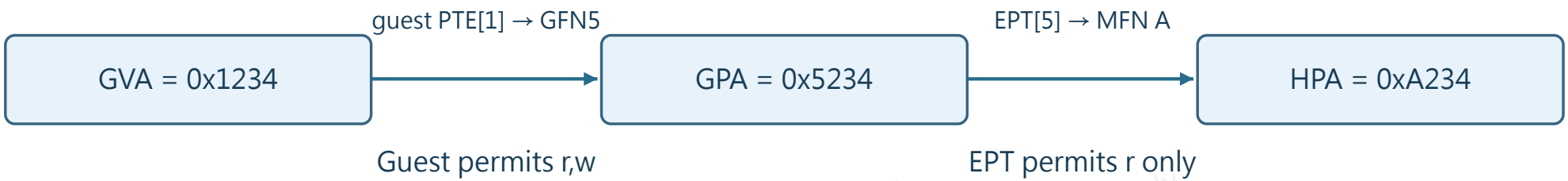
Root/non-root or host/guest context adds an axis beyond user/kernel. Guest kernel can run non-root ring 0, app non-root ring 3, while VMM manages root context. VM entry loads guest state; configured events exit, save state/provide reason, and let VMM handle/resume.

Architecture/controls determine exits; not every instruction or syscall exits. Fidelity/control require correct intercepts, state retention, event injection. IOMMU and interrupt remapping isolate/route device activity separately from CPU virtualization. EPT alone does not constrain all DMA.

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Shadow tables and hardware two-stage translation

Compose addresses and permissions at both stages



4096 B pages preserve offset 0x234 at both stages. Read passes; second stage denies write.

Shadow maps VPN1 → MFN A directly, requiring synchronized guest PTE updates and invalidation.

Guest tables map GVA→GPA and VMM maps GPA→HPA. Hardware two-stage EPT/NPT combines both during walking. A shadow table is a VMM-maintained GVA→HPA composition used by an ordinary MMU, requiring interception/tracking of guest table changes. Extra tables do not make them one algorithm.

Shadow tables and hardware two-stage translation (continued)

With4096 B pages, GVA0x1234=(VPN1, offset 0x234); guestPTE[1]=(GFN5, rw) gives GPA0x5234. EPT[5]=(MFN0xA, r) gives HPA0xA234. Read needs permission at both; EPT denies write despite guest w. Guest-table denial follows guest fault semantics; second-stage violation uses VMM handling. They are not one identical fault.

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Why uncached nested walks are not twice as expensive

Assume m guest levels, n second-stage levels, resident page tables and data, valid permissions, and no TLB/walk caches or large pages. Each guest PTE read requires n second-stage PTE reads plus its own read, totaling m(n+1). Translating the final data GPA adds n. Thus the number of **table reads** is m(n+1)+n=(m+1)(n+1)−1; add one data read.

For m=n=4 this is 24 table reads plus one data read, not 4+4. It is a cold model, not the cost of every load; TLB hits, walk caches and huge pages can reduce actual reads. Shadow tables have a shorter hardware walk but require VMM maintenance after guest updates.

The count excludes writes: assume accessed/dirty bits need no updates. Real hardware may additionally write those bits. Mapping and permission changes require architecture-specific translation-cache invalidation; correct tagging or flushing prevents reuse of another VM's mapping.

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Trace a two-stage page walk

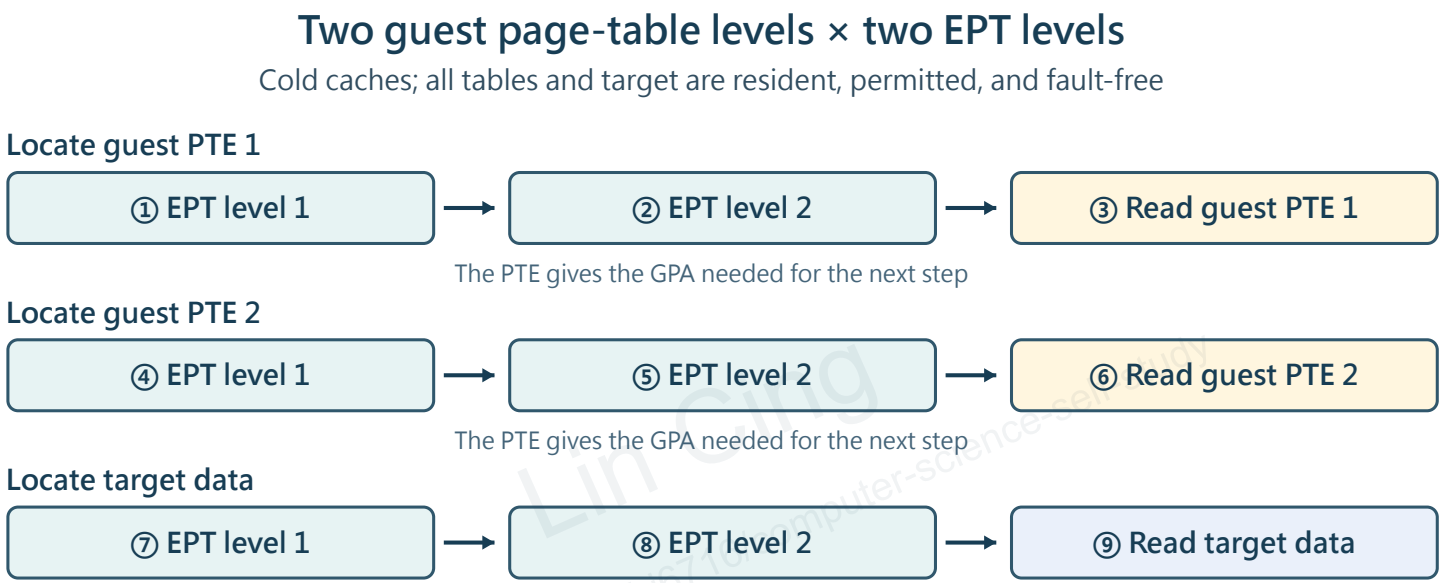


Table reads: 2 + 1 + 2 + 1 + 2 = 8; data reads: 1. A/D-bit updates and other writes are excluded.

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Trace a two-stage page walk (continued)

With two guest levels, two EPT levels, cold caches, resident pages and valid permissions, each guest PTE GPA needs two EPT reads before the PTE read. The final data GPA needs two more EPT reads before the data read. The total is eight table reads and one data read; accessed/dirty-bit updates and other writes are excluded.

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Lifecycle and type responsibilities

Creation validates CPU, memory, disk, network needs and policy, allocates vCPUs, address spaces, devices, then loads the guest. Running VMs may pause, resume, snapshot, migrate. Deletion first stops devices/events before resource reclamation; deleting management metadata while DMA/network identity remains active is insufficient.

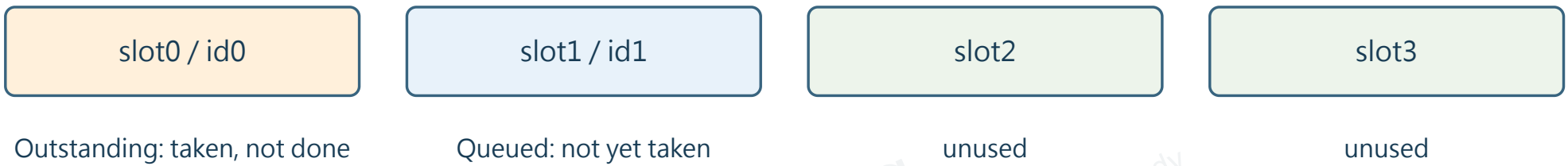
Type0 partitions via firmware/hardware with dedicated/shared resources. Type1 directly manages host resources, including VMMs in some general-purpose kernels. Type2 uses host process/device/memory services, often with privileged helpers/drivers and hardware assistance. Type2 does not mean no host awareness or no VT-x.

Types do not universally rank performance/functionality. Nesting depends on VMM, CPU, configuration, not solely type 0. Compare management scope, TCB, and resource paths.

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Paravirtual I/O and a four-slot ring

Four slots, ordered completion: backend consumption does not free a slot



req_prod=2 · req_cons=1 · rsp_prod=0 · rsp_cons=0 ; used=2 · free=4−2=2 。

Complete id0: rsp_prod→1. Frontend consumes response, rsp_cons→1; only then release the slot.

Historical reference is Xen SOSP2003 cooperation, hypercalls, shared rings. This simplified single-producer/consumer model is not every Xen ABI. Counters increase monotonically; slot=index mod4. With req_prod2, req_cons1, rsp_prod0, rsp_cons0, queued=1, outstanding=1, response=0, used=2, free=2.

Paravirtual I/O and a four-slot ring (continued)

Backend completing id0 advances rsp_prod1, making outstanding0, response1. Frontend consumes it, advances rsp_cons1, leaving used=req_prod−rsp_cons=1, free3. Backend taking a request does not permit descriptor reuse while completion is pending.

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Ring synchronization and validation

Write each descriptor before release-publishing req_prod; the consumer acquire-observes prod before reading it. Write a response before publishing rsp_prod; the frontend consumes it before advancing rsp_cons to release the slot. A doorbell is notification, not a substitute for the counter predicate.

A recheck alone can lose a wakeup: consumer observes empty, producer publishes and notifies before the consumer registers, then consumer sleeps. In a teaching lock/condition model, the producer publishes the predicate and wakes under the same lock. The consumer holds that lock, checks in a while loop, then **atomically registers and releases the lock while sleeping**; it rechecks after waking. Where domains cannot share this lock, an event-count protocol must ensure that publication in the check-to-sleep window remains observable; ordinary notifications alone do not suffice.

With nonwrapping counters and ordered completion, $0 \le \text{req_prod} - \text{rsp_cons} \le N$ and $\text{rsp_cons} \le \text{rsp_prod} \le \text{req_cons} \le \text{req_prod}$ keep the four regions disjoint. Backend validates ID, operation, block range, length and buffer grants; shared memory does not authorize arbitrary guest DMA. Out-of-order completion requires a different ID/slot model.

Process VMs, emulation, and containment

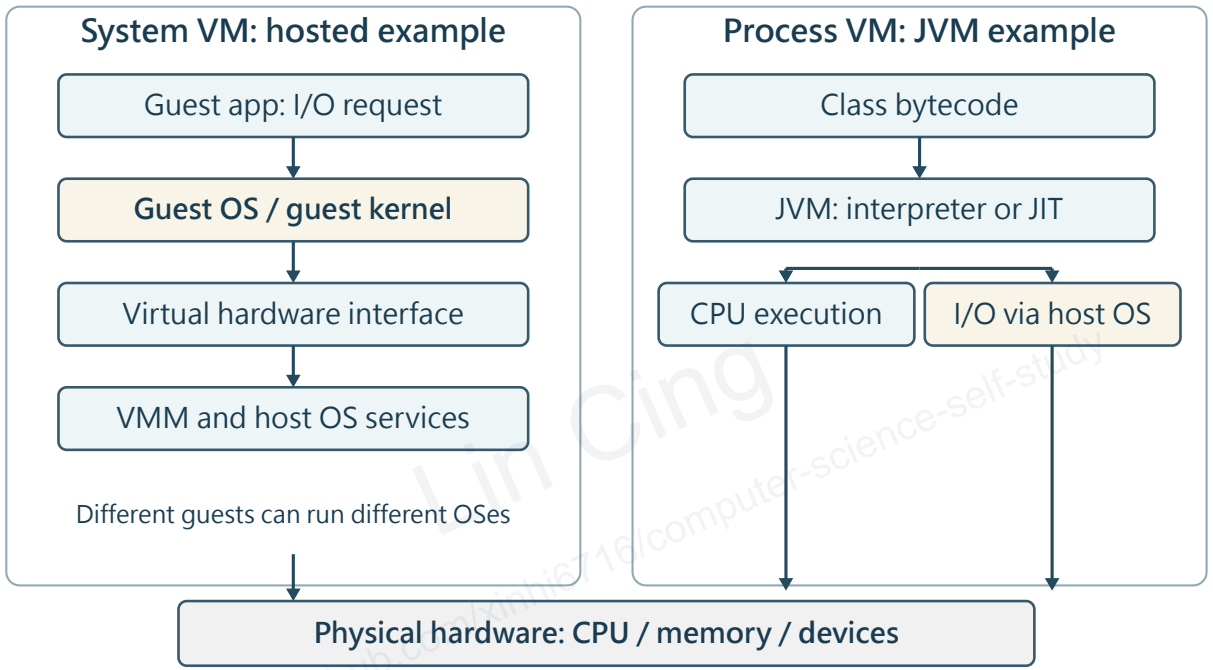
JVM/.NET expose bytecode/runtime rather than a full guest OS. Cross-ISA emulation interprets/translates instructions/machine state to run incompatible binaries. Neither is classified solely by owning hardware ring 0.

Solaris10 zones/BSD jails share the host kernel while separating namespaces, resources, and security contexts. Different userland/network views do not create independent kernels. Shared-kernel failure differs from VM isolation. Solaris global zone manages non-global zones; services/resource controls need configuration rather than an independent kernel scheduler per zone.

Testing another kernel needs a system VM/emulator; separating same-kernel services can use containment; bytecode needs its runtime. Choose an appropriate interface rather than a universal winner.

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System VM and JVM: interfaces and kernels



Hosted example: isolation needs management and resource controls.

A system VM exposes virtual hardware to a guest OS. A JVM runs bytecode and still uses the host kernel. The host OS path shown handles services such as I/O; not every instruction is a system call. Isolation also depends on the VMM and management plane, and shared CPU, memory, and I/O can cause interference.

Two scheduling layers

Two pCPUs serve A 0/A 1 and B0/B1, ignoring host overhead. A runs0–10 ms, B10–20, A 20–30, B30–40. A thread needing20 ms CPU on A 0 completes at 30; B's counterpart at 40. Equal long-run halves do not make every20 ms budget have40 ms response.

Guest-ready threads wait while their vCPU lacks pCPU time. Descheduling a guest lock holder can waste another vCPU's spinning allocation. Separate virtual clock, wall clock, vCPU runtime. Expired virtual timers may await vCPU scheduling. Guest priorities alone cannot provide host-level real-time admission/isolation/scheduling guarantees.

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Balloon direction and capacity conservation

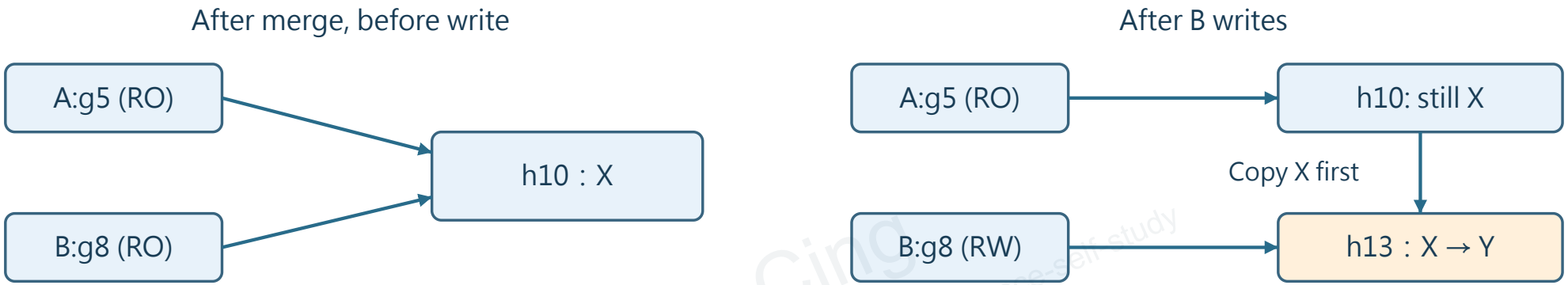
In Waldspurger's ESX OSDI 2002 design, a guest balloon driver reserves guest pages and reports PFNs; the VMM can reclaim their host backing while guest policy selects other pages to reclaim or swap. Inflation reduces workload-usable guest memory; it creates no RAM.

A 4096 MiB host reserves 512, leaving 3584 for guests. A and B each demand 2304 MiB of resident backing: 4608 total, not yet fully backed, exceeds availability by 1024. Balloon A by 256 and B by 768 to target A=2048 and B=1536, totaling 3584; adding VMM 512 gives 4096. This conserves demand and allocation targets, not an impossible prior allocation of 4608 MiB of guest RAM. Nominal guest sizes can stay fixed while usable backing changes at a cost.

Balloon pages are reserved by the driver and should not be used by the normal guest workload. For this teaching isolation model, assume an erroneous access to an unbacked mapping faults, so the VMM rejects it or installs safe backing; this is not a universal promise about every balloon protocol. Deflation must restore safe use; slow or uncooperative drivers may require other paging or limits.

Sharing, COW, and double paging

Compare bytes before sharing; split host frames before writing



B write fault: allocate/copy → change B mapping and invalidate → allow write. A stays unchanged.

A:g5→h10 and B:g8→h12 contain X. A hash selects candidates. Lock the state, write-protect both pages, invalidate writable TLB/EPT entries, and exclude DMA writes. Under this protection compare all bytes and map equal pages to read-only h10. Reclaim h12 only after references, pins, and in-flight accesses are gone.

Sharing, COW, and double paging (continued)

B's write causes a COW fault: copy X to private h13, change B:g8→h13, invalidate old translations as required, then permit the write of Y; A still sees X. Guest and VMM swapping can cause double paging. Ballooning may improve victim selection; scanning and synchronization have costs, and sharing must respect isolation policy.

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I/O paths and virtual networking

Emulation validates virtual-register operations before host-driver service. Paravirtual devices use cooperative queues. Passthrough lets a device serve a guest directly, still requiring IOMMU isolation, interrupt routing, and safe reset/ownership transfer.

If DMA0x5000 maps to host 0xA000, IOMMU must confine the permitted range. CPU EPT alone is not DMA confinement. Passthrough may require extra device-state support for snapshots/migration; RAM copying alone is insufficient.

Bridging joins an L2 network. NAT maps internal addresses/ports to external tuples. A 10.0.2.15:5000 and B10.0.2.16:5000 can use host:40001/40002 and share one public IP. Firewalls still define guest-to-guest/external channels.

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Storage mappings and snapshot chains

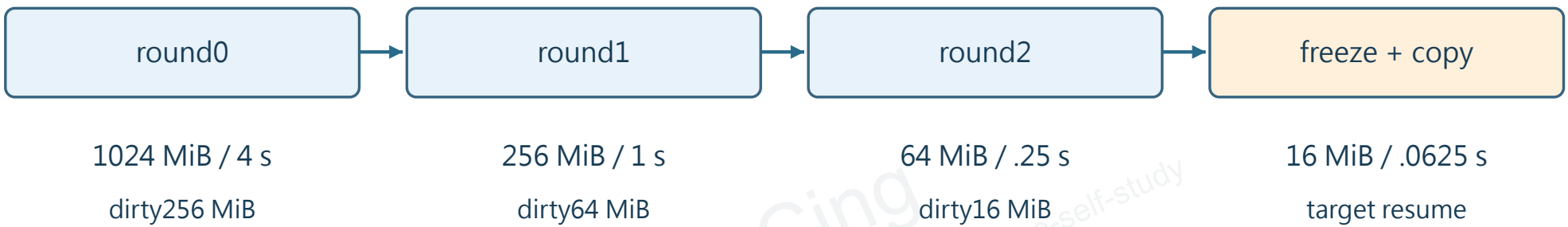
Guest LBA maps through an image/host filesystem to storage, or directly to a block volume; a virtual disk need not be one ordinary file. Thin provisioning postpones allocation without guaranteeing all logical capacities can be filled together.

Base={0:A,1:B,2:C}; after snapshot, active delta={1:B′}. Read1 finds B′ in delta, read 2 falls through to C. Selecting base restores B at 1 but does not retract sent mail. A shared base cannot be deleted while clones reference it. Backups need the required chain, metadata, and consistent point.

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Pre-copy migration round by round

Normal pre-copy: total time differs from downtime



B=256 MiB/s, d=64 MiB/s. Total 1360 MiB / 5.3125 s; pause .0625 s, excluding control cost.

Assume compatible VMM/CPU, shared storage, no unmigratable device, effective copy B=256 MiB/s, distinct dirtied pages d=64 MiB/s, and ignore protocol/CPU-state cost. R0=1024 copies in 4 s, creating R1=256; next1 s creates R2=64; next0.25 s creates R3=16. Freeze and copy16 in 0.0625 s. Total1360 MiB over5.3125 s; pause0.0625 s.

Pre-copy migration round by round (continued)

Generally $R(i+1)=(d/B)R(i)$, geometrically shrinking if $d/B<1$. At $d \geq B$ it does not shrink; cap rounds, reduce dirtying, extend pause, or change method. Repeated dirty pages, variable bandwidth, and finite RAM make reality richer than this explicit model.

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Stop-and-copy and one active owner

Freeze includes state-changing DMA/devices/events, not only user code. Save vCPU/device/dirty RAM and required network state. Source must stop externally executing the guest before ownership transfer; reclaim source after target operation is established.

If target starts but acknowledgment is lost, source restarting can create split brain. A safe protocol needs explicit commit/ownership and failure recovery. Waiting for acknowledgment alone is not a proof of all failure cases. This simplified normal path proves a consistent handoff, not transparent availability under arbitrary partitions.

Maintain external address/MAC/routing reachability. TCP tolerating a short pause does not make downtime zero. Shared storage is this model's premise; block migration exists elsewhere, so it does not prove all live migration excludes disk copying.

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Workstation, Xen, and ESX

The book's circa2013 hosted Workstation diagram routes guest app→guest kernel→virtual device/CPU→VMM/host services→hardware. Ordinary instructions can execute directly; I/O can use host services. It is an architecture example, not a release's complete implementation or measurement.

Xen SOSP2003 uses cooperative guests/hypercalls/shared rings. ESX OSDI2002 manages overcommit with ballooning/content sharing. Their periods/interfaces differ. Early Xen PV is not every later Xen mode; one paper benchmark does not establish a universal performance ratio.

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Two required properties

Fidelity preserves specified guest-visible transitions; resource control prevents escaping CPU, address, device management. Traps, translation, hypercalls, hardware assistance implement these properties rather than prove themselves.

Permission intersection, ring publication/reuse, COW-before-write, and single-owner migration keep guest-visible and host-managed state consistent. A successful demonstration does not cover every transition.

VCPU count is not pCPU capacity; nominal RAM is not current host backing; logical disk size is not allocated bytes; snapshots do not roll back the outside world; migration transfer time is not downtime. Identify the layer of every number. Exercises reconstruct modes, addresses, walk counts, ring counters, balloon capacity, COW, and migration recurrences.

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References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz et al., Operating System Concepts, 9th ed., Ch. 16. Xen SOSP 2003. ESX memory management OSDI 2002. JVM SE7 verifier.

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